

Olympiad Test Paper — Mathematics (Class 7) — Paper 3

Chapter 9: Geometric Twins

Paper 3 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	9 — Geometric Twins
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Beware of treating SSA or AAA as a congruence rule, confusing congruence with mere similarity, miscounting the order of rotational symmetry, and forgetting that the equal angle in SAS must be the included one. Do not write on this paper — mark answers on your answer sheet.

1. In two triangles ABC and PQR you are told $AB = PQ$, $BC = QR$ and $\angle C = \angle R$. A student declares them congruent by SAS. Is the student right, and why or why not?

- (A) Yes — two sides and an angle always give SAS, so the triangles are congruent.
- (B) No — the equal angle (at C and R) is NOT the angle included between the two pairs of equal sides, so this is the ambiguous SSA case and congruence is not guaranteed.
- (C) No, but they are congruent by ASA instead, using the two sides as if they were angles.
- (D) Yes — the triangles are congruent by SSS because two sides already match and the third is then forced.

2. Triangle DEF is congruent to triangle XYZ written in that exact letter order. If $DE = 5$ cm, $EF = 7$ cm, $FD = 9$ cm, which side of triangle XYZ measures 7 cm?

- (A) XY
- (B) YZ
- (C) ZX
- (D) It cannot be decided from the letter order.

3. Two triangles satisfy: $\angle A = \angle P = 50^\circ$, $\angle B = \angle Q = 60^\circ$, and $AB = PQ = 6$ cm. By which criterion are triangles ABC and PQR congruent, and is the side AB correctly positioned for it?

- (A) AAA — three angles match once you find the third, so they are congruent by AAA.
- (B) ASA — AB lies between the two equal angles A and B, so it is the included side, making the triangles congruent.
- (C) SAS — two angles and a side count as two sides and an angle.
- (D) They are only similar, not congruent, because the third angle is the same in both.

4. How many lines of symmetry does a non-square rhombus have?

- (A) 2 — its two diagonals.
- (B) 4 — the same as a square.
- (C) 1 — only the longer diagonal.
- (D) 0 — a rhombus has no line of symmetry.

5. What is the order of rotational symmetry of a regular pentagon?

- (A) 10
- (B) 1
- (C) 2
- (D) 5

6. In triangle ABC, $AB = AC$ and AD is the bisector of angle A meeting BC at D. Using the congruence of the two halves, which statement about BD, DC and angle ADB is fully correct?

- (A) $BD = DC$ only; angle ADB need not be 90° .
- (B) angle $ADB = 90^\circ$ only; BD and DC need not be equal.
- (C) Neither follows, because the two halves are merely similar, not congruent.
- (D) $BD = DC$ and angle $ADB = 90^\circ$, because triangles ABD and ACD are congruent by SAS ($AB = AC$, angle $BAD = \text{angle } CAD$, AD common).

7. Two arcs of fixed radii are used to construct a triangle on a given base. Why do exactly the side-lengths 3 cm, 4 cm, 5 cm produce only ONE triangle (up to congruence), whereas being told only two of the three sides would not?

- (A) With all three sides fixed, SSS forces a unique triangle; with only two sides the two arcs can meet at more than one point or at none of fixed shape, leaving the triangle undetermined.
- (B) Because 3-4-5 is a right triangle, and only right triangles are uniquely fixed by their sides.
- (C) Because 3 cm, 4 cm, 5 cm have no common factor, and coprime sides always give a unique triangle.
- (D) Two sides actually do fix a unique triangle; the third side adds no new information.

8. A figure looks the same after being reflected in a vertical line through its centre AND after a 180° rotation about its centre. Which capital letter (standard block form) fits BOTH descriptions?

- (A) A
- (B) S
- (C) H
- (D) Z

9. A scalene triangle (all three sides of different lengths) has how many lines of symmetry?

- (A) 1
(C) 2
- (B) 3
(D) 0

10. A regular hexagon is rotated about its centre. Through what is the SMALLEST positive angle of rotation that maps it onto itself, and what is its order of rotational symmetry?

- (A) 60° , order 6.
(C) 60° , order 12.
- (B) 120° , order 3.
(D) 30° , order 12.

11. Triangles ABC and DEF have $AB = DE = 6$ cm, $AC = DF = 6$ cm, and $BC = EF = 6$ cm. A student says 'they are congruent by SAS.' What is the most precise correct statement?

- (A) They are congruent by SAS, since equal sides also force equal included angles.
(B) They are congruent, but by SSS (three equal sides), not SAS — no angle was given.
(C) They are only similar, because equal sides give equal shape but the size could differ.
(D) Congruence cannot be claimed because no angle is given at all.

12. In triangle PQR, $PQ = PR$ and the bisector of angle P meets QR at S. Combining symmetry and congruence, which set of conclusions is entirely correct?

- (A) $QS = SR$, but PS need not be perpendicular to QR.
(B) angle PSQ = 90° , but QS and SR need not be equal.
(C) PS bisects QR only if angle P happens to be 90° .
(D) PS is the perpendicular bisector of QR, $QS = SR$, and angle PSQ = 90° .

13. Which of these has EXACTLY one line of symmetry?

- (A) A rectangle (non-square).
(C) An isosceles (non-equilateral) triangle.
- (B) An equilateral triangle.
(D) A circle.

14. If triangle ABC is congruent to triangle DEF and the perimeter of ABC is 30 cm with $AB = 8$ cm and $BC = 11$ cm, what is the length of side DF?

- (A) 8 cm
(C) 11 cm
- (B) 30 cm
(D) Cannot be found without the angles.

15. Two right triangles have a right angle each. In the first, the legs are 9 cm and 12 cm. In the second, the hypotenuse is 15 cm and one leg is 9 cm. Are they congruent, and by what reasoning?

- (A) No — the second triangle's other leg is unknown, so congruence cannot be settled.
(B) Yes, congruent by SSA, since two sides and a non-included angle match.
(C) Yes, congruent by RHS — the first has hypotenuse $\sqrt{9^2+12^2}=15$ cm, so both share a 15 cm hypotenuse and a 9 cm leg with the right angle.
(D) Only similar, because one is described by legs and the other by hypotenuse.

16. A square is rotated about its centre. How many times in one full 360° turn (counting the final return to start) does it look exactly the same as its starting position?

- (A) 8
(C) 2
- (B) 4
(D) 1

17. When a figure is reflected across a line, which of the following is ALWAYS true of the image compared with the original?

- (A) It is similar but larger than the original.
- (B) It is congruent to the original (same size and shape) but its orientation (left/right sense) is reversed.
- (C) It is congruent and keeps the same orientation as the original.
- (D) Its area is doubled, since two copies now exist.

18. Two triangles ABC and DEF have angle A = angle D = 90° . Also $AB = DE = 5$ cm and $AC = DF = 12$ cm (the two legs). Which criterion most directly proves congruence, and is RHS appropriate here?

- (A) RHS, because both triangles contain a right angle.
- (B) ASA, since the right angle counts as one of two equal angles.
- (C) SAS is the most direct (two legs with the included right angle equal); RHS is not needed because we have both legs, not the hypotenuse.
- (D) AAA, because both have a 90° angle and the shapes match.

19. A figure has rotational symmetry of order 1. What does this tell you?

- (A) It has NO rotational symmetry beyond the trivial full turn — only a 360° rotation brings it back, so effectively it is not rotationally symmetric.
- (B) It looks the same after a 90° rotation.
- (C) It has exactly one line of symmetry.
- (D) It can be turned through 180° and look unchanged.

20. In triangles ABC and DEF, $AB = DE$, angle A = angle D, and angle B = angle E. A student concludes congruence by ASA. A second student says it is actually AAS for some pairing. Which is the safest correct verdict?

- (A) Not congruent — two angles and a side never fix a triangle.
- (B) Only similar, since two equal angles give the same shape but the size is free.
- (C) Congruent by SAS, treating the two angles as the 'two sides'.
- (D) Congruent by ASA — AB is the side included between the two equal angles A and B.

21. Point P is reflected across a line m to give image P'. Which property of the segment PP' relative to line m is correct?

- (A) Line m passes through P' but not through the midpoint of PP'.
- (B) Line m is parallel to PP'.
- (C) P and P' are the same distance from m only if PP' is horizontal.
- (D) Line m is the perpendicular bisector of PP' — it crosses PP' at its midpoint at a right angle.

22. If triangle ABC $\cong$ triangle DEF, angle A = 48° and angle E = 67° , what is the measure of angle C?

- (A) 65°
- (B) 67°
- (C) 48°
- (D) Cannot be found.

23. Which English capital letter (block form) has rotational symmetry of order 2 but NO line of symmetry?

- (A) O
- (B) T
- (C) S
- (D) E

24. A triangle has two sides 6 cm and 8 cm with a 5 cm side, matched against another triangle with sides 6 cm, 8 cm and 5 cm. A child argues 'they need not be congruent because no angle is given.' Evaluate this.

- (A) The child is wrong — all three pairs of sides are equal, so SSS proves them congruent without any angle.
- (B) The child is right — at least one angle is always required to prove congruence.
- (C) They are only similar, because three sides fix shape but not size.
- (D) They are congruent, but only by SAS using the 6 cm and 8 cm sides.

25. An equilateral triangle is reflected in one of its three lines of symmetry. The image is then rotated 120° about the triangle's centre. The final figure compared with the original equilateral triangle is:

- (A) Larger than the original because two transformations were applied.
- (B) A mirror image that can never be made to coincide with the original.
- (C) Rotated 240° from the original and so out of place.
- (D) Congruent and exactly coincident with the original (occupies the same region), because both a reflection in an axis and a 120° rotation map an equilateral triangle onto itself.

26. Triangles ABC and ABD share side AB, with C and D on opposite sides of line AB. Given $AC = AD$ and $BC = BD$, by which criterion is triangle ABC $\cong$ triangle ABD?

- (A) SSS — $AC = AD$, $BC = BD$, and AB is common to both.
- (B) SAS, using AB and the angle at A.
- (C) RHS, because the two triangles together look like a kite with a right angle.
- (D) ASA, using the two base angles.

27. How many lines of symmetry and what order of rotational symmetry does a regular octagon have?

- (A) 8 lines of symmetry but rotational symmetry of order 4.
- (B) 8 lines of symmetry and rotational symmetry of order 8.
- (C) 4 lines of symmetry and rotational symmetry of order 8.
- (D) 16 lines of symmetry and rotational symmetry of order 8.

28. A rectangle that is NOT a square has how many lines of symmetry, and what is its order of rotational symmetry?

- (A) 2 lines of symmetry and rotational symmetry of order 2.
- (B) 4 lines of symmetry and rotational symmetry of order 4.

- (C) 2 lines of symmetry and rotational symmetry of order 1.
- (D) 0 lines of symmetry and rotational symmetry of order 2.

29. In triangle ABC, $AB = AC$ and angle $A = 36^\circ$. The angle bisector from B meets AC at D. Using base-angle and isosceles reasoning, what is angle DBC?

- (A) 72°
- (B) 36°
- (C) 18°
- (D) 54°

30. Two congruent figures are placed so that one is the mirror image of the other across a line. Which statement about their areas and perimeters is correct?

- (A) Perimeters are equal but areas may differ.
- (B) Areas are equal but perimeters may differ.
- (C) Both their areas and their perimeters are equal, because reflection preserves all lengths and the enclosed region's size.
- (D) Both area and perimeter are halved in the mirror image.

31. If triangle ABC $\cong$ triangle PQR, which of the following equalities is guaranteed by CPCT (corresponding parts)?

- (A) angle B = angle Q and $AC = PR$.
- (B) angle B = angle R and $AC = PQ$.
- (C) $AB = QR$ and angle A = angle Q.
- (D) angle C = angle P and $BC = PR$.

32. An equilateral triangle has rotational symmetry of order 3. After how many degrees of the SMALLEST positive rotation does it first coincide with itself, and how does this relate to its three lines of symmetry?

- (A) 60° , with 3 lines of symmetry — half the rotation gives the same look.
- (B) 120° ; it has order 3 rotation AND 3 lines of symmetry, the counts matching because it is a regular polygon.
- (C) 120° , but only 1 line of symmetry through the top vertex.
- (D) 90° , with 4 lines of symmetry like a square.

33. Which statement is FALSE?

- (A) If two triangles have equal areas, they must be congruent.
- (B) If two triangles are congruent, they are also similar.
- (C) AAA proves similarity but not congruence.
- (D) Reflection produces a figure congruent to the original.

34. A capital letter 'N' is reflected in a vertical mirror line. The result looks identical to which letter?

- (A) A reversed N, which looks like the letter 'И' / a back-to-front N — it does NOT match a normal N (because N has no vertical line of symmetry).
- (B) A normal N, because N is symmetric about a vertical line.
- (C) The letter Z, because reflection turns N into Z.
- (D) The letter M, because N and M are reflections of each other.

35. In triangle ABC, the perpendicular from A to BC has foot M, and you observe $BM = MC$. What can you conclude, and by which congruence does it follow?

- (A) Nothing can be concluded; the foot being the midpoint is a coincidence.
- (B) angle $A = 90^\circ$, by RHS.
- (C) $AB = AC$ (the triangle is isosceles), since triangles ABM and ACM are congruent by SAS ($BM = MC$, the right angles at M equal, AM common).
- (D) The triangle is equilateral, by SSS.

36. Two triangles satisfy angle $B = \text{angle } Q = 90^\circ$, hypotenuse $AC = PR = 13 \text{ cm}$, and one leg $AB = PQ = 5 \text{ cm}$. Which criterion proves congruence, and why is this NOT the ambiguous SSA case?

- (A) SSA — and so the triangles need not be congruent.
- (B) SAS, with the right angle included between the hypotenuse and the leg.
- (C) RHS proves congruence; the right angle fixes the configuration so the third side is uniquely 12 cm, removing the SSA ambiguity.
- (D) ASA, using the right angle and one other angle.

37. Which figure has rotational symmetry of order MORE than 1 but ZERO lines of symmetry?

- (A) A non-square rectangle.
- (B) An isosceles trapezium.
- (C) A regular pentagon.
- (D) A parallelogram that is neither a rectangle nor a rhombus.

38. A figure is mapped onto itself by a rotation of 90° about a point. What is the LEAST possible order of its rotational symmetry?

- (A) 2
- (B) 90
- (C) 4
- (D) 1

39. In triangle ABC, $AB = AC = 10 \text{ cm}$ and $BC = 12 \text{ cm}$. The altitude from A meets BC at M. Using the isosceles symmetry, what is the length AM?

- (A) 6 cm
- (B) 10 cm
- (C) 8 cm
- (D) 2 cm

40. Two triangles have all three angles respectively equal (AAA), and additionally one pair of corresponding sides are equal. What can be concluded?

- (A) Only similar, because AAA can never give congruence even with a side.
- (B) They are congruent — once one pair of corresponding sides is equal, equal angles (ASA/AAS) fix the size as well.
- (C) Congruent by AAA, because all three angles match.
- (D) Nothing, because mixing angle and side conditions is not allowed.

41. The diagonal of a kite ABCD (with $AB = AD$ and $CB = CD$) is AC. Reflecting the kite across line AC maps B onto:

- (A) C, the far vertex.
- (B) D — line AC is the axis of symmetry of the kite, swapping B and D.
- (C) A, the near vertex.
- (D) the midpoint of BD.

42. Triangle ABC is isosceles with $AB = AC$. We write the self-correspondence $ABC \leftrightarrow ACB$. This 'maps the triangle to itself' and shows angle B = angle C. Which congruence criterion justifies this self-congruence?

- (A) SAS — $AB = AC$, angle A = angle A (common), $AC = AB$, mapping the triangle onto its mirror image, giving angle B = angle C by CPCT.
- (B) SSS, because all three sides obviously match a triangle with itself.
- (C) ASA, using angle B and angle C which we are trying to prove equal.
- (D) RHS, because the triangle must contain a right angle.

43. A figure looks the same after rotations of 90° , 180° and 270° . How many lines of symmetry MUST it have?

- (A) Exactly 4.
- (B) Exactly 2.
- (C) It need not have any — order-4 rotational symmetry does not by itself guarantee a line of symmetry.
- (D) At least 1.

44. In triangle ABC and triangle DEF, $AB = DE = 7$ cm, angle B = angle E = 40° , and $BC = EF = 9$ cm. The student must pick the criterion. The correct one is:

- (A) SSA — the angle is not between the two sides, so congruence is uncertain.
- (B) SAS — the equal angle B (= E) is included between the equal sides AB and BC.
- (C) ASA — two equal angles and a side.
- (D) RHS — the 40° angle plays the role of a right angle.

45. Which of these regular figures has the GREATEST order of rotational symmetry: equilateral triangle, square, regular hexagon, regular dodecagon (12 sides)?

- (A) The regular hexagon, with order 6.
- (B) The square, because four right angles maximise symmetry.
- (C) All four are equal, since every regular polygon has the same rotational order.
- (D) The regular dodecagon, with order 12.

46. Triangle ABC has angle A = angle B. A line is drawn so that triangle ABC reflects onto itself. Which line is the axis of symmetry?

- (A) The perpendicular bisector of AB (equivalently, the line through C and the midpoint of AB), because angle A = angle B makes $CA = CB$, so the triangle is isosceles with axis through C.
- (B) The line AB itself.
- (C) The perpendicular bisector of CA.
- (D) There is no axis of symmetry.

47. Two triangles ABC and DEF have $AB = DE$, $BC = EF$ and a pair of equal angles. For congruence to be GUARANTEED, the equal angle must be:

- (A) the angle at A and D (opposite sides BC and EF) -- a side-side-angle match still pins the triangle down.
- (B) the angle at C and F -- then SSA still guarantees congruence.
- (C) the angle at B and E (between AB and BC, and between DE and EF) — the included angle, giving SAS.
- (D) any one of the three pairs of angles; the position does not matter.

48. A pattern is built from a single motif repeated by rotating it through 72° steps about a centre, with no reflection. What is the order of rotational symmetry, and how many lines of symmetry?

- (A) Order 5 and 5 lines of symmetry.
- (B) Order 72 and 0 lines of symmetry.
- (C) Order 5 and 1 line of symmetry.
- (D) Order 5 rotational symmetry and 0 lines of symmetry.

49. In triangle DEF, $DE = DF$ and angle $D = 3 \cdot \text{angle } E$. Using isosceles symmetry, the measure of angle E is:

- (A) 36°
- (B) 45°
- (C) 60°
- (D) 30°

50. Which set of three measurements does NOT determine a unique triangle (i.e. fails to guarantee congruence among triangles sharing them)?

- (A) Three sides (SSS).
- (B) Two angles and the included side (ASA).
- (C) Two sides and the included angle (SAS).
- (D) Two sides and a non-included angle (SSA).

51. A semicircle (half-disc, including its straight diameter edge) has how many lines of symmetry and what order of rotational symmetry?

- (A) Infinite lines of symmetry, like a full circle.
- (B) 2 lines of symmetry and order 2.
- (C) 1 line of symmetry and order 2.
- (D) 1 line of symmetry and rotational symmetry of order 1 (none).

52. Two triangles are reflections of each other across a line (mirror images). A student insists 'a triangle and its mirror image can never be congruent because one is left-handed and one is right-handed.' Evaluate this for triangles.

- (A) The student is right — mirror images are only similar, never congruent.
- (B) The student is right for scalene triangles but wrong for isosceles ones.
- (C) Mirror images are congruent only if the triangle is equilateral.
- (D) The student is wrong — a triangle and its reflection are always congruent (all corresponding sides and angles are equal); congruence allows flips, so 'handedness' does not block it.

- 53.** Triangle $ABC \cong$ triangle DEF , and you know angle $A = 2 \cdot$ angle B , angle $C = 80^\circ$. What is angle E (in triangle DEF)?
- (A) 66.7° , the value of angle A .
(B) About 33.3° (one-third of 100°), since angle $A +$ angle $B = 100^\circ$, angle $A = 2 \cdot$ angle B gives angle $B = 100/3 \approx 33.3^\circ$, and E corresponds to B .
(C) 80° , the value of angle C . (D) 50° , half of 100° .
- 54.** Which statement about lines of symmetry is TRUE?
- (A) Every quadrilateral has at least one line of symmetry.
(B) A scalene triangle has exactly one line of symmetry.
(C) A circle has infinitely many lines of symmetry, since any diameter is an axis.
(D) An angle (two rays from a vertex) has no line of symmetry.
- 55.** In triangle ABC , AD is the median to BC and it is also perpendicular to BC . What does this combination force, and by which congruence?
- (A) angle $A = 90^\circ$, by RHS.
(B) $AB = AC$, the triangle is isosceles, by SAS on triangles ABD and ACD ($BD = DC$ since AD is a median, equal right angles at D , AD common).
(C) Nothing extra — a median can always be perpendicular without consequence.
(D) The triangle is equilateral, by SSS.
- 56.** An equilateral triangle and a regular hexagon are compared. Which has more lines of symmetry, and by how many more?
- (A) They have the same number of lines of symmetry.
(B) The equilateral triangle has more, since fewer sides means tighter symmetry.
(C) The regular hexagon, with 6 lines, has 3 more than the equilateral triangle's 3 lines.
(D) The hexagon has 6 more lines than the triangle.
- 57.** Two triangles ABC and PQR have angle $A =$ angle P , $AB = PQ$, and $AC = PR$. A student writes 'congruent by ASA.' What is the correct criterion, and is the student right?
- (A) ASA is correct, treating the two sides as the two angles.
(B) SSA — and so congruence is not guaranteed.
(C) AAA, because the third angle will also match.
(D) SAS — angle A is included between sides AB and AC ; the student is wrong to call it ASA because only one angle is given.
- 58.** A figure has both a horizontal and a vertical line of symmetry. What is the MINIMUM order of rotational symmetry it must have?
- (A) 1 — two mirror lines do not force any rotational symmetry.
(B) 4 — two mirror lines always give order-4 rotation.
(C) 0 — rotational order is unrelated to mirror lines.
(D) 2 — having two perpendicular mirror lines forces at least order-2 (180°) rotational symmetry.

59. Triangle $ABC \cong$ triangle BAC for the SAME triangle ABC (note the swapped first two letters). What does this self-congruence force?

- (A) $AB = BC$, making it isosceles with apex A .
- (B) $CA = CB$, so the triangle is isosceles with the apex at C (angle $A =$ angle B).
- (C) The triangle is equilateral.
- (D) angle $C = 90^\circ$.

60. Consider these claims: (i) every figure with a line of symmetry also has rotational symmetry of order > 1 ; (ii) every figure with rotational symmetry of order > 1 also has a line of symmetry. Which is/are FALSE?

- (A) Both (i) and (ii) are false.
- (B) Only (i) is false.
- (C) Only (ii) is false.
- (D) Both are true.

Solutions — Mathematics (Class 7), Chapter 9: Geometric Twins — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	B	21	D	31	A	41	B	51	D
2	B	12	D	22	A	32	B	42	A	52	D
3	B	13	C	23	C	33	A	43	C	53	B
4	A	14	C	24	A	34	A	44	B	54	C
5	D	15	C	25	D	35	C	45	D	55	B
6	D	16	B	26	A	36	C	46	A	56	C
7	A	17	B	27	B	37	D	47	C	57	D
8	C	18	C	28	A	38	C	48	D	58	D
9	D	19	A	29	B	39	C	49	A	59	B
10	A	20	D	30	C	40	B	50	D	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) For SAS the equal angle must sit between the two equal sides. Here AB and BC meet at B, but the given equal angle is at C — it is opposite side AB, not included between AB and BC. So the data is Side-Side-Angle (SSA), which is the ambiguous case: two non-congruent triangles can fit the same SSA data. Calling it SAS ignores the word 'included.' ASA needs two angles, but only one angle is given, so that path fails too. SSS would need the third side to be forced, but SSA does not force the third side uniquely. Hence congruence cannot be claimed.

2. (B) Congruence written as $DEF \cong XYZ$ matches vertices in order: $D \leftrightarrow X$, $E \leftrightarrow Y$, $F \leftrightarrow Z$. So side EF corresponds to side YZ. Since $EF = 7$ cm, $YZ = 7$ cm. DE corresponds to XY (= 5) and FD to ZX (= 9), so those are wrong. The letter order is exactly what tells us the correspondence, so 'cannot be decided' is false.

3. (B) Angles A and B are at the ends of side AB, so AB is the side included between the two equal angles. Two angles and the included side give ASA, which guarantees

congruence. AAA never proves congruence (it only gives similarity), so even though the third angle is also 70° in both, that alone would not be enough — what makes it work is the equal included side. SAS is mislabelled here; we have two angles, not two sides.

4. (A) A rhombus has all four sides equal but its angles are not all 90° (otherwise it would be a square). Its two diagonals are its axes of reflection: folding along either diagonal maps the figure onto itself. The mid-lines joining midpoints of opposite sides are NOT axes for a rhombus (they are for a rectangle/square but not a rhombus), so it is not 4. It clearly has more than one and certainly more than zero.

5. (D) A regular n -gon has rotational symmetry of order n : it coincides with itself after each rotation of $360^\circ/n$. For a regular pentagon, $360^\circ/5 = 72^\circ$, and there are exactly 5 positions (at 72° , 144° , 216° , 288° , 360°) where it maps onto itself, so the order is 5. The figure also has 5 lines of symmetry, but the question asks for rotational order, not total symmetries (which would tempt 10). Order 1 means no rotational symmetry, which is false for a regular polygon.

6. (D) Compare triangles ABD and ACD: $AB = AC$ (given), angle BAD = angle CAD (AD bisects A), and AD is common. That is two sides and the included angle, so SAS gives congruence. By CPCT, $BD = DC$ (so D is the midpoint) and angle ADB = angle ADC; since these two add to the straight angle 180° , each is 90° . So BOTH results hold. They are genuinely congruent (same size), not just similar, because AD is shared, fixing the size.

7. (A) SSS states three pairs of equal sides guarantee congruence — the triangle's shape and size are then fully determined, so any 3-4-5 triangle is congruent to any other. With only two sides given, the third vertex is not pinned down: the included angle can vary, giving infinitely many different triangles, so two sides are not enough. The uniqueness has nothing to do with the triangle being right-angled (every SSS triple is unique, right-angled or not) nor with the sides being coprime.

8. (C) We need a letter with both a vertical line of symmetry and 180° rotational symmetry. H has a vertical axis (left/right mirror) and also looks identical after a half-turn, so it satisfies both. A has a vertical line of symmetry but no rotational symmetry (upside-down it is not an A). S and Z have 180° rotational symmetry but no line of symmetry (their mirror image is reversed). Only H meets both conditions.

9. (D) A line of symmetry would fold the triangle onto itself, which requires at least two equal sides on either side of the fold. In a scalene triangle all three sides differ, so no fold maps it onto itself — it has 0 lines of symmetry. An isosceles triangle has exactly 1 and an equilateral has 3, but scalene has none.

10. (A) A regular hexagon has 6 equal sides, so its rotational order is 6. The smallest mapping rotation is $360^\circ/6 = 60^\circ$. After every 60° it lands on itself, giving 6 distinct positions in a full turn. Order 3 with 120° would be true for a figure that only repeats every

third of a turn (like an equilateral triangle pattern), not a regular hexagon. There is no order 12 for a plain regular hexagon.

11. (B) All three pairs of sides are equal, which is precisely the SSS criterion, so the triangles ARE congruent — and being all 6 cm, both are equilateral. The student's reasoning is wrong because SAS needs a stated included angle; here we cite SSS. Equal sides do force the triangle (so it is not 'only similar' — congruent triangles are the same size), and no angle is needed for SSS, so the last option is also wrong.

12. (D) Triangles PQS and PRS: $PQ = PR$, angle $QPS = \text{angle } RPS$ (PS bisects P), PS common — SAS, so they are congruent. By CPCT, $QS = SR$ and angle $PSQ = \text{angle } PSR$; these are equal and sum to 180° , so each is 90° . Thus PS is both the median and the altitude — i.e. the perpendicular bisector of QR. All three conclusions hold simultaneously, regardless of the size of angle P (the isosceles condition alone is enough, not a right angle at P).

13. (C) An isosceles triangle that is not equilateral has exactly one axis: the line through the apex and the midpoint of the base. A non-square rectangle has 2 axes (the two mid-lines), an equilateral triangle has 3, and a circle has infinitely many. So only the isosceles triangle has exactly one.

14. (C) From the perimeter, $AC = 30 - 8 - 11 = 11$ cm. Under $ABC \cong DEF$ the correspondence is $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$, so DF corresponds to AC (both join the first and third vertices). Hence $DF = AC = 11$ cm. DF is not AB ($=8$, that corresponds to DE) and is certainly not the whole perimeter. The angles are not needed — corresponding sides are equal under congruence.

15. (C) In the first triangle the hypotenuse is $\sqrt{81+144} = \sqrt{225} = 15$ cm. So both triangles have a right angle, a hypotenuse of 15 cm, and a leg of 9 cm — exactly the RHS criterion (Right angle, Hypotenuse, Side). RHS guarantees congruence. The second's other leg is in fact forced to be 12 cm by Pythagoras, so it is not unknown. RHS is a valid rule and is NOT the same as the ambiguous SSA — the right angle plus hypotenuse removes the ambiguity. Being congruent, they are more than just similar.

16. (B) A square has rotational symmetry of order 4: at 90° , 180° , 270° and 360° it coincides with its start. That is 4 matching positions in a full turn. The number 8 confuses rotational order (4) with total symmetries including its 4 reflection lines ($4+4=8$), but the question asks only about rotation. So the answer is 4.

17. (B) A reflection is a rigid motion (an isometry): it preserves all lengths and angles, so the image is congruent to the original. However a reflection flips orientation — a left hand reflects to a right hand. It does not enlarge anything (that would be wrong about size and similarity), it does not keep the same orientation (rotation/translation do, reflection does not), and it certainly does not change the area of the figure itself.

18. (C) The right angle at A (and D) is included between the two legs AB and AC (DE and DF). Two sides and the included angle is exactly SAS, the most direct rule here. RHS is for when you know the hypotenuse and one leg plus the right angle — but here both legs are given, so SAS is cleaner (RHS would still work after computing the hypotenuse, but it is not the most direct). ASA fails because only one angle is given, and AAA never proves congruence.

19. (A) Order of rotational symmetry counts how many positions in a full turn map the figure onto itself. Order 1 means only the complete 360° turn works — i.e. the figure has no genuine rotational symmetry. It does not look the same at 90° (that would be order 4) or at 180° (that would be order 2 or more). Rotational order is unrelated to the COUNT of mirror lines, so 'exactly one line of symmetry' is a different property.

20. (D) Angles A and B sit at the two ends of side AB, so AB is the included side between the two equal angles — this is exactly ASA, which guarantees congruence. Two angles plus a side DO fix a triangle (the third angle is forced, and a side fixes the size), so the second option is false. It is not 'only similar': the equal side fixes the size, making them congruent. SAS is the wrong label because we have angles, not two sides.

21. (D) By definition of reflection, the mirror line is the perpendicular bisector of the segment joining a point to its image. So m meets PP' at its midpoint and at 90° , and P, P' are equidistant from m regardless of orientation. The mirror line is not parallel to PP' (it is perpendicular to it), and it does not pass through P' itself unless P were on the line. Equal distance from the mirror always holds, not only for horizontal segments.

22. (A) Under $ABC \cong DEF$, angle B corresponds to angle E, so angle B = 67° . The three angles of triangle ABC sum to 180° , so angle C = $180^\circ - 48^\circ - 67^\circ = 65^\circ$. Angle C equals angle F by correspondence, but its value comes from the angle sum. 67° is angle B, 48° is angle A, so neither is C, and it certainly can be found.

23. (C) S looks the same after a 180° turn (order 2 rotational symmetry) but has no mirror line — its reflection is reversed. O has infinite rotational symmetry and many lines of symmetry, so it fails the 'no line' condition. T has one vertical line of symmetry but no rotational symmetry. E has a horizontal line of symmetry but no rotational symmetry. Only S has order-2 rotation with no mirror line.

24. (A) SSS says three pairs of equal sides are by themselves enough to guarantee congruence — no angle needed. So the child's premise is mistaken. Three equal sides fix BOTH shape and size (each angle is determined by the sides via the triangle's rigidity), so 'only similar' is wrong. It is SSS, not SAS, because we are matching three sides, not two-sides-and-an-included-angle.

25. (D) Both transformations are symmetries of the equilateral triangle: a reflection in any of its three axes maps it onto itself, and a 120° rotation (since order = 3) also maps it

onto itself. Applying two such symmetries still leaves the triangle covering its original region — the result coincides with the original. Rigid motions never change size, so it is not larger. Because each step is a self-symmetry, the figure does coincide (not a stuck mirror image, not displaced).

26. (A) List the three pairs: $AC = AD$ (given), $BC = BD$ (given), and $AB = AB$ (shared side). That is three pairs of equal sides — SSS — which proves congruence. No angle is stated, so SAS and ASA cannot be invoked directly. There is no given right angle, so RHS does not apply; the kite shape does not force a right angle here.

27. (B) For a regular n -gon, both the number of lines of symmetry and the order of rotational symmetry equal n . A regular octagon ($n = 8$) therefore has 8 lines of symmetry and rotational order 8 (smallest turn $360^\circ/8 = 45^\circ$). The other options mismatch the two counts, but for a regular polygon they are always equal to each other and to n .

28. (A) A non-square rectangle has 2 lines of symmetry — the two lines joining midpoints of opposite sides (the diagonals are NOT axes for a non-square rectangle). It also has rotational symmetry of order 2: a 180° turn maps it onto itself. A square would have 4 of each, but a plain rectangle does not. Order 1 would mean no rotational symmetry, which is false since the half-turn works.

29. (B) Since $AB = AC$, base angles B and C are equal; each $= (180 - 36)/2 = 72^\circ$. BD bisects angle B (72°), so angle DBC $= 36^\circ$. Mistaking the full base angle gives 72° ; halving 36° wrongly gives 18° ; 54° comes from an unrelated computation. The correct value is half of 72° , i.e. 36° .

30. (C) Reflection is an isometry: every length is preserved, so the perimeter (a sum of lengths) is unchanged, and the enclosed area is unchanged too. Congruent figures always share equal area and equal perimeter; a mirror image is still congruent. Nothing is halved or altered — only the left/right orientation flips.

31. (A) The correspondence $A \leftrightarrow P$, $B \leftrightarrow Q$, $C \leftrightarrow R$ fixes every matching part. So angle B = angle Q (not angle R), and side AC (joining A and C) corresponds to side PR (joining P and R), giving $AC = PR$. The other options mismatch the order: AB corresponds to PQ (not QR), angle A to angle P (not Q), angle C to angle R (not P), BC to QR (not PR). Only the first option respects the correspondence throughout.

32. (B) Order 3 means it coincides at $360^\circ/3 = 120^\circ$ (and 240° , 360°). 60° would map it to an upside-down position that is NOT the same triangle, so 60° is wrong. A regular polygon's lines of symmetry equal its rotational order, so an equilateral triangle has 3 lines (one through each vertex and the opposite midpoint), not 1. A square's $90^\circ/4$ figures are irrelevant here.

33. (A) Two triangles can have the same area while having completely different shapes (e.g. a tall thin triangle and a short wide one), so equal area does NOT force congruence —

that statement is false. The others are true: congruent triangles are a special case of similar (with ratio 1); AAA fixes shape only (similarity), not size; and reflection is an isometry, giving a congruent image.

34. (A) N has no vertical line of symmetry, so reflecting it in a vertical mirror produces a reversed N (the diagonal stroke slants the other way). It does not stay a normal N (that would require a vertical axis it lacks). A vertical reflection does not turn N into Z (that is a different transformation) nor into M (M and N are not mirror images of each other). N does, however, have 180° rotational symmetry — a different property from the vertical reflection asked about here.

35. (C) In triangles ABM and ACM: $BM = MC$ (given), $\angle AMB = \angle AMC = 90^\circ$ ($AM \perp BC$), and AM is common. Two sides and the included right angle give SAS, so the triangles are congruent. By CPCT, $AB = AC$, so the triangle is isosceles. The midpoint condition is not coincidental — it forces the isosceles property. There is no reason angle A is 90° , and equilateral would require all three sides equal, which is not shown.

36. (C) We have a Right angle, equal Hypotenuses, and an equal Side (leg) — exactly RHS, which guarantees congruence. Although the given angle is not between the two given sides (so superficially it resembles SSA), the special fact that it is a RIGHT angle, paired with the hypotenuse, kills the ambiguity: Pythagoras forces the other leg to be $\sqrt{13^2 - 5^2} = 12$ cm in both. The right angle is not the included angle between the hypotenuse and the leg (those two sides meet at an acute vertex), so SAS is mislabelled; ASA needs two angles, which are not given.

37. (D) A general parallelogram has rotational symmetry of order 2 (a 180° turn maps it onto itself) but no line of symmetry (its diagonals and mid-lines do not fold it onto itself). A non-square rectangle has 2 lines of symmetry, so it fails the 'zero lines' test; an isosceles trapezium has 1 line of symmetry; a regular pentagon has 5 lines. Only the general parallelogram pairs rotational symmetry with no axis.

38. (C) If a 90° rotation is a symmetry, then repeating it gives symmetries at 180° , 270° and 360° as well — four positions in a full turn — so the order is at least 4. It cannot be 2 (180° only), because a 90° symmetry forces order divisible by 4. Order 90 is meaningless here (order counts positions, and $360/90 = 4$). Order 1 means no symmetry, contradicting the given 90° symmetry.

39. (C) The altitude from the apex of an isosceles triangle is also the perpendicular bisector of the base (by the SAS congruence of the two halves), so $BM = MC = 6$ cm. In right triangle ABM, $AM = \sqrt{AB^2 - BM^2} = \sqrt{100 - 36} = \sqrt{64} = 8$ cm. The value 6 cm is BM, not AM; 10 is the slant side AB; 2 comes from a wrong subtraction. The altitude is 8 cm.

40. (B) AAA alone gives only similarity. But the moment ONE pair of corresponding sides is equal, two equal angles plus a corresponding side become ASA (or AAS), which proves congruence — the equal side locks the scale. So they ARE congruent. It is the equal side, not 'AAA', that does the work, so 'congruent by AAA' is the wrong justification even though the conclusion of congruence is right. Combining angle and side data is perfectly legitimate.

41. (B) A kite with $AB = AD$ and $CB = CD$ is symmetric about the diagonal joining the vertices between the equal-length pairs — that is AC. Reflection in AC therefore swaps the two off-axis vertices B and D, so B maps to D. A and C lie on the axis and stay fixed (B cannot map to a fixed vertex), and B does not map to the midpoint of BD (reflection maps a vertex to a vertex here).

42. (A) The correspondence $ABC \leftrightarrow ACB$ pairs AB with AC, the common angle A with itself, and AC with AB — two sides and the included angle, i.e. SAS. By CPCT this yields angle B = angle C, the base-angle theorem. ASA would illegitimately assume the very equality (angle B = angle C) we are trying to prove. While 'a triangle matches itself' is trivially true side-for-side, the elegant proof of equal base angles uses the SAS self-map, not a trivial SSS, and no right angle is involved so RHS does not apply.

43. (C) Rotational symmetry and reflection symmetry are independent. A figure such as a pinwheel or an S-shaped four-armed swastika-like motif can have rotational order 4 yet zero lines of symmetry. So order-4 rotation does NOT force any mirror line. (A square happens to have 4 lines too, but that is a property of the square, not a consequence of order-4 rotation.) Hence the guaranteed minimum number of lines is zero.

44. (B) Sides AB and BC meet at vertex B, and the equal angle is at B (and E). So the angle IS included between the two equal sides — this is SAS, which guarantees congruence. It is not SSA: SSA would be when the equal angle is NOT between the two equal sides, but here it is. ASA requires two equal angles (only one is given). RHS requires a right angle and a hypotenuse, neither stated.

45. (D) A regular n-gon has rotational order n. So the orders are: triangle 3, square 4, hexagon 6, dodecagon 12. The largest is the dodecagon at 12. The hexagon (6) is smaller, the square's right angles do not maximise anything, and regular polygons do NOT all share the same order — order grows with the number of sides.

46. (A) Equal base angles A and B force the opposite sides equal: $CA = CB$ (converse of the isosceles base-angle result). So the triangle is isosceles with apex C, and its axis of symmetry passes through C and the midpoint of AB, i.e. it is the perpendicular bisector of AB. Side AB is a side, not an axis; the perpendicular bisector of CA is not an axis here; and since the triangle is isosceles it certainly does have an axis.

47. (C) AB and BC are included at vertex B (DE and EF at E). For SAS — the only rule available with two sides and one angle — the equal angle must be that included angle at B/E. If instead the equal angle were at A/D or C/F, it would be a non-included angle, giving SSA, which does NOT guarantee congruence. So the position very much matters; only the included angle works.

48. (D) Repeating every 72° means $360^\circ/72^\circ = 5$ positions, so rotational order is 5. Because the construction uses rotation only and no reflection, the motif need not be mirror-symmetric, so the whole pattern has no line of symmetry — 0 lines. Order is the count of positions (5), not the angle (so 'order 72' is wrong). Rotational symmetry alone does not create mirror lines, so 5 or 1 lines are wrong.

49. (A) $DE = DF$ means base angles E and F are equal: angle E = angle F. With angle D = 3·angle E, the angle sum gives $3E + E + E = 180^\circ$, so $5E = 180^\circ$, $E = 36^\circ$. Then angle F = 36° and angle D = 108° . Choosing 45° (from $4E = 180$) miscounts the equal base angles; 60° and 30° come from other miscounts. The correct base angle is 36° .

50. (D) SSA — two sides and an angle that is not between them — is the ambiguous case: two non-congruent triangles can share the same SSA data, so it does not determine a unique triangle. SSS, ASA and SAS each fix a triangle uniquely (they are the valid congruence criteria). Hence SSA is the odd one out.

51. (D) A semicircle has a single axis: the perpendicular to the diameter through its midpoint (folding the curved part onto itself). It has NO rotational symmetry other than the trivial 360° (order 1) — a 180° turn would put the flat edge on top, not matching. It is not like a full circle (infinite symmetry) because the straight edge breaks most symmetry. A half-turn does not map it onto itself, so order is 1, not 2.

52. (D) Reflection is an isometry: it preserves every length and angle, so a triangle and its mirror image have all corresponding parts equal — they are congruent. Congruence permits reflections (you may flip a figure to superimpose it), so left/right 'handedness' is irrelevant. This holds for ALL triangles, not just isosceles or equilateral ones; an isosceles or equilateral triangle additionally coincides with its own reflection by symmetry, but even a scalene triangle is congruent to its mirror image.

53. (B) Angles A and B sum to $180^\circ - 80^\circ = 100^\circ$. With angle A = 2·angle B, we get $2B + B = 100^\circ$, so $3B = 100^\circ$, $B = 100/3 \approx 33.3^\circ$. Under $ABC \cong DEF$, angle E corresponds to angle B, so angle E $\approx 33.3^\circ$. The value 66.7° is angle A (twice B), 80° is angle C, and 50° would be the answer only if A equalled B, which it does not. Correspondence sends B to E, giving $\approx 33.3^\circ$.

54. (C) Any diameter of a circle folds the circle exactly onto itself, and there are infinitely many diameters, so a circle has infinitely many lines of symmetry — true. A general (irregular) quadrilateral need have none, so 'every quadrilateral' is false. A scalene triangle

has zero, not one, lines of symmetry. An angle DOES have a line of symmetry: its bisector folds one ray onto the other.

55. (B) AD a median gives $BD = DC$; AD perpendicular to BC gives angle $ADB = \text{angle } ADC = 90^\circ$; AD is common. Triangles ABD and ACD therefore match by SAS (two sides and the included right angle), so $AB = AC$ by CPCT — the triangle is isosceles. This says nothing about angle A being 90° , and it is more than 'no consequence' — a median that is also an altitude forces the isosceles property. Equilateral would need all three sides equal, which is not implied.

56. (C) A regular n-gon has n lines of symmetry: the equilateral triangle has 3, the regular hexagon has 6. So the hexagon has $6 - 3 = 3$ more. They are not equal, fewer sides does not give more symmetry, and the difference is 3, not 6 (6 is the hexagon's total, not the difference).

57. (D) Sides AB and AC meet at vertex A, and the given equal angle is at A — so it is the included angle. Two sides with the included angle is SAS, which guarantees congruence. The student's 'ASA' is wrong because ASA requires two angles, but only one angle is given. It is not SSA (the angle IS between the two sides), and AAA is irrelevant since we are not relying on angles to match.

58. (D) Composing the reflection in the horizontal axis with the reflection in the vertical axis (which are perpendicular and meet at a point) produces a rotation of 180° about that point — a symmetry of the figure. So the figure must have at least order-2 rotational symmetry. It need not be order 4 (that would require diagonal mirrors too, as in a square). Order 1 or 0 would mean no rotational symmetry, contradicting the forced 180° symmetry.

59. (B) The correspondence $ABC \leftrightarrow BAC$ pairs $A \leftrightarrow B$, $B \leftrightarrow A$, $C \leftrightarrow C$. So side CA (from C to A) corresponds to side CB (from C to B), forcing $CA = CB$ — the triangle is isosceles with apex C, and the base angles A and B are equal. It does NOT make $AB = BC$ (that would be a different correspondence), and the swap of just two vertices does not force all three sides equal (equilateral) nor a right angle at C.

60. (A) Reflection symmetry and rotational symmetry are independent. (i) is false: an isosceles (non-equilateral) triangle has 1 line of symmetry but only order-1 rotation (no genuine rotational symmetry). (ii) is false: a general parallelogram has order-2 rotational symmetry but no line of symmetry. Since each property can occur without the other, both claims fail. So neither implication holds — both are false.